ON DATABASE **mynewpaltz**

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on s.ssn = e.ssn group by s.
ssn, s.firstName, s.lastName having count(e.courseID) = 5;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Kevin     | Green    | 100000041 |
| Michelle  | Adams    | 100000042 |
| Brian     | Nelson   | 100000043 |
| Carol     | Baker    | 100000044 |
| Edward    | Hall     | 100000045 |
| Ronald    | Campbell | 100000047 |
| Melissa   | Mitchell | 100000048 |
| Timothy   | Carter   | 100000049 |
| Deborah   | Roberts  | 100000050 |
| Stephanie | Phillips | 100000052 |
| Jeffrey   | Evans    | 100000053 |
| Rebecca   | Turner   | 100000054 |
| Ryan      | Diaz     | 100000055 |
| Laura     | Parker   | 100000056 |
| Sharon    | Edwards  | 100000058 |
| Gary      | Collins  | 100000059 |
| Cynthia   | Reyes    | 100000060 |
+-----------+----------+-----------+
17 rows in set (0.013 sec)
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on s.ssn = e.ssn group by s.
ssn, s.firstName, s.lastName having count(e.courseID) = 4;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Amanda    | Rivera   | 100000046 |
| Jacob     | Cruz     | 100000057 |
+-----------+----------+-----------+
2 rows in set (0.001 sec)
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on s.ssn = e.ssn group by s.
ssn, s.firstName, s.lastName having count(e.courseID) = 3;
+-------------+-----------+-----------+
| firstName   | lastName  | ssn       |
+-------------+-----------+-----------+
| James       | Smith     | 100000001 |
| Mary        | Johnson   | 100000002 |
| Robert      | Williams  | 100000003 |
| Patricia    | Brown     | 100000004 |
| John        | Jones     | 100000005 |
| Jennifer    | Garcia    | 100000006 |
| Michael     | Miller    | 100000007 |
| Linda       | Davis     | 100000008 |
| David       | Rodriguez | 100000009 |
| Elizabeth   | Martinez  | 100000010 |
| William     | Hernandez | 100000011 |
| Barbara     | Lopez     | 100000012 |
| Richard     | Gonzalez  | 100000013 |
| Susan       | Wilson    | 100000014 |
| Joseph      | Anderson  | 100000015 |
| Jessica     | Thomas    | 100000016 |
| Thomas      | Taylor    | 100000017 |
| Sarah       | Moore     | 100000018 |
| Christopher | Jackson   | 100000019 |
| Karen       | Martin    | 100000020 |
+-------------+-----------+-----------+
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on s.ssn = e.ssn group by s.
ssn, s.firstName, s.lastName having count(e.courseID) = 2;
+-----------+------------+-----------+
| firstName | lastName   | ssn       |
+-----------+------------+-----------+
| Nicholas  | Stewart    | 100000061 |
| Kathleen  | Morris     | 100000062 |
| Eric      | Morales    | 100000063 |
| Amy       | Murphy     | 100000064 |
| Stephen   | Cook       | 100000065 |
| Shirley   | Rogers     | 100000066 |
| Jonathan  | Gutierrez  | 100000067 |
| Angela    | Ortiz      | 100000068 |
| Larry     | Morgan     | 100000069 |
| Helen     | Cooper     | 100000070 |
| Justin    | Peterson   | 100000071 |
| Anna      | Bailey     | 100000072 |
| Scott     | Reed       | 100000073 |
| Brenda    | Kelly      | 100000074 |
| Brandon   | Howard     | 100000075 |
| Pamela    | Ramos      | 100000076 |
| Frank     | Kim        | 100000077 |
| Nicole    | Foster     | 100000078 |
| Benjamin  | Alvarado   | 100000079 |
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on s.ssn = e.ssn group by s.
ssn, s.firstName, s.lastName having count(e.courseID) = 1;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Charles   | Lee      | 100000021 |
| Nancy     | Perez    | 100000022 |
| Matthew   | Thompson | 100000023 |
| Lisa      | White    | 100000024 |
| Anthony   | Harris   | 100000025 |
| Betty     | Sanchez  | 100000026 |
| Mark      | Clark    | 100000027 |
| Margaret  | Ramirez  | 100000028 |
| Donald    | Lewis    | 100000029 |
| Sandra    | Robinson | 100000030 |
| Steven    | Walker   | 100000031 |
| Ashley    | Young    | 100000032 |
| Paul      | Allen    | 100000033 |
| Dorothy   | King     | 100000034 |
| Andrew    | Wright   | 100000035 |
| Kimberly  | Scott    | 100000036 |
| Joshua    | Torres   | 100000037 |
| Emily     | Nguyen   | 100000038 |
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s left join EnrollmentTable e on s.ssn = e.ssn where
e.courseID is null;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Jason     | Gomez    | 100000051 |
| Jack      | Myers    | 100000091 |
+-----------+----------+-----------+
2 rows in set (0.003 sec)
```

```
MariaDB [mynewpaltz]> select subjectId, count(courseID) as num_courses from CourseTable where numOfCredits = 4 group by subjectId;
+-----------+-------------+
| subjectId | num_courses |
+-----------+-------------+
| CS        |           2 |
| ENGR      |           2 |
| MATH      |           2 |
+-----------+-------------+
3 rows in set (0.003 sec)
```

```
MariaDB [mynewpaltz]> select subjectId, count(courseID) as num_courses from CourseTable where numOfCredits = 3 group by subjectId;
+-----------+-------------+
| subjectId | num_courses |
+-----------+-------------+
| CS        |           3 |
| ENGL      |           5 |
| ENGR      |           2 |
| MATH      |           3 |
+-----------+-------------+
4 rows in set (0.001 sec)
```

```
MariaDB [mynewpaltz]> select subjectId, count(courseID) as num_courses from CourseTable where numOfCredits = 2 group by subjectId;
+-----------+-------------+
| subjectId | num_courses |
+-----------+-------------+
| ENGR      |           1 |
+-----------+-------------+
1 row in set (0.001 sec)
```

```
MariaDB [mynewpaltz]> select * from CourseTable where numOfCredits = 5;
Empty set (0.010 sec)

MariaDB [mynewpaltz]> select * from CourseTable where numOfCredits = 1;
Empty set (0.000 sec)

MariaDB [mynewpaltz]>
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, c.title, e.grade from StudentTable s join EnrollmentTable e on s.ssn = e.ssn
join CourseTable c on e.courseID = c.courseID;
+-------------+------------+---------------------------+-------+
| firstName   | lastName   | title                     | grade |
+-------------+------------+---------------------------+-------+
| James       | Smith      | Calculus I                | A     |
| Jennifer    | Garcia     | Calculus I                | B     |
| William     | Hernandez  | Calculus I                | B     |
| Jessica     | Thomas     | Calculus I                | C     |
| Charles     | Lee        | Calculus I                | A     |
| Kevin       | Green      | Calculus I                | A     |
| Edward      | Hall       | Calculus I                | B     |
| Timothy     | Carter     | Calculus I                | A     |
| Cynthia     | Reyes      | Calculus I                | A     |
| Anna        | Bailey     | Calculus I                | B     |
| Frank       | Kim        | Calculus I                | C     |
| Janet       | Ford       | Calculus I                | A     |
| Tyler       | Woods      | Calculus I                | A     |
| Mary        | Johnson    | Calculus II               | B     |
```

```
MariaDB [mynewpaltz]> select c.courseNumber, c.subjectId, count(e.ssn) as enrolled_students from CourseTable c join EnrollmentTable
 e on c.courseID = e.courseID where c.courseID = 10 group by c.courseID, c.courseNumber, c.subjectId;
+--------------+-----------+-------------------+
| courseNumber | subjectId | enrolled_students |
+--------------+-----------+-------------------+
|          450 | CS        |                13 |
+--------------+-----------+-------------------+
1 row in set (0.010 sec)
```

```
MariaDB [mynewpaltz]> select courseID, count(ssn) as num_students from EnrollmentTable group by courseID;
+----------+---------------+
| courseID | num_students |
+----------+---------------+
|        1 |            13 |
|        2 |            13 |
|        3 |            12 |
|        4 |            14 |
|        5 |            13 |
|        6 |            11 |
|        7 |            13 |
|        8 |            13 |
|        9 |            12 |
|       10 |            13 |
|       11 |            14 |
|       12 |            13 |
|       13 |            14 |
|       14 |            14 |
|       15 |            12 |
|       16 |            12 |
|       17 |            12 |
|       18 |            12 |
|       19 |            11 |
|       20 |            10 |
+----------+---------------+
20 rows in set (0.009 sec)
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, sum(c.numOfCredits) as total_credits from StudentTable s join EnrollmentTable
 e on s.ssn = e.ssn join CourseTable c on e.courseID = c.courseID group by s.ssn, s.firstName, s.lastName;
+-------------+------------+---------------+
| firstName   | lastName   | total_credits |
+-------------+------------+---------------+
| James       | Smith      |            11 |
| Mary        | Johnson    |            10 |
| Robert      | Williams   |            10 |
| Patricia    | Brown      |             9 |
| John        | Jones      |             9 |
| Jennifer    | Garcia     |            10 |
| Michael     | Miller     |            10 |
| Linda       | Davis      |            10 |
| David       | Rodriguez  |            10 |
| Elizabeth   | Martinez   |            10 |
| William     | Hernandez  |             9 |
| Barbara     | Lopez      |            10 |
| Richard     | Gonzalez   |             9 |
| Susan       | Wilson     |            10 |
| Joseph      | Anderson   |            10 |
| Jessica     | Thomas     |            11 |
| Thomas      | Taylor     |            10 |
| Sarah       | Moore      |            10 |
| Christopher | Jackson    |             9 |
```

ON DATABASE **nations**

```
MariaDB [nation]> select c.name as continent, count(r.region_id) as number_of_regions from continents c join regions r o
n c.continent_id = r.continent_id group by c.name;
+---------------+-------------------+
| continent     | number_of_regions |
+---------------+-------------------+
| Africa        |                 5 |
| Antarctica    |                 1 |
| Asia          |                 4 |
| Europe        |                 6 |
| North America |                 3 |
| Oceania       |                 5 |
| South America |                 1 |
+---------------+-------------------+
7 rows in set (0.002 sec)
```

```
MariaDB [nation]> select c.name as continent, r.name as region from continents c join regions r on c.continent_id = r.co
ntinent_id order by c.name, r.name;
+---------------+---------------------------+
| continent     | region                    |
+---------------+---------------------------+
| Africa        | Central Africa            |
| Africa        | Eastern Africa            |
| Africa        | Northern Africa           |
| Africa        | Southern Africa           |
| Africa        | Western Africa            |
| Antarctica    | Antarctica                |
| Asia          | Eastern Asia              |
| Asia          | Middle East               |
| Asia          | Southeast Asia            |
| Asia          | Southern and Central Asia |
| Europe        | Baltic Countries          |
| Europe        | British Islands           |
| Europe        | Eastern Europe            |
| Europe        | Nordic Countries          |
| Europe        | Southern Europe           |
| Europe        | Western Europe            |
| North America | Caribbean                 |
| North America | Central America           |
| North America | North America             |
| Oceania       | Australia and New Zealand |
```

```
MariaDB [nation]> select r.name as region from regions r group by r.name having count(distinct r.conti
nent_id) > 1;
Empty set (0.003 sec)
```

```
MariaDB [nation]> create view ContriesAndContinents as select co.name as country, c.name as continent from countries co join region
s r on co.region_id = r.region_id join continents c on r.continent_id = c.continent_id order by co.name asc;
Query OK, 0 rows affected (0.014 sec)

MariaDB [nation]> select * from ContriesAndContinents;
+--------------------------------------------+---------------+
| country                                    | continent     |
+--------------------------------------------+---------------+
| Afghanistan                                | Asia          |
| Albania                                    | Europe        |
| Algeria                                    | Africa        |
| American Samoa                             | Oceania       |
| Andorra                                    | Europe        |
| Angola                                     | Africa        |
| Anguilla                                   | North America |
| Antarctica                                 | Antarctica    |
| Antigua and Barbuda                        | North America |
| Argentina                                  | South America |
| Armenia                                    | Asia          |
| Aruba                                      | North America |
| Australia                                  | Oceania       |
| Austria                                    | Europe        |
| Azerbaijan                                 | Asia          |
```